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## **A New Tool for Performance Monitoring of Pseudo Dry Gas Technology Validated with Experimental Data**

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### **Abstract**

Remote deepwater gas field development poses challenges due to the long tieback and requirement for remote control and monitoring. Recently, a novel Pseudo Dry Gas (inline gas-liquid separator) technology has been developed and extensively tested in a 6" flow loop at low and high pressure, to tackle the long tieback issue. The technology offers significantly lower CAPEX and OPEX by removing liquid phases from the main gas line, hence, reducing the back pressure and flow assurance issue related to wax or hydrate formation.

Given the harsh environment and inaccessibility of deepwater installations, a tool is required for remote monitoring (in real-time) and prediction of the performance of this novel Pseudo Dry Gas (PDG) system and to help the operator to optimize production. The paper aims to present the development of the solution. The tool is able to predict PDG performance efficiency, multiphase flow behavior, and associated FA issues in the flowline based on continuous input from field sensors. In this work, a transient numerical model of the PDG technology has been developed and combined with a transient multiphase flow simulator using a plug-in based architecture.

The tool was validated with the extensive 6" flow loop data and tested under dynamic conditions covering ramp-up or turn-down, shutdown, normal operation, and operation beyond the design envelope. The flow loop test data covers low and high pressure, horizontal and near horizontal, and various fluids representing gas, water, condensate, oil, and oil-water mixture. In addition to that, a steady state case comparison with a commercial software was conducted for a deepwater field.

The validation results show the ability of the tool to predict PDG performance (the separation and liquid level in the liquid chamber) under the dynamic condition with good accuracy. The dynamic test results showed that PDG technology effectively separates most of the liquid within its operating envelope producing a pseudo-dry gas. The liquid removal reduced pressure drop in the main gas line by half,

minimizing slugging, and liquid surge to receiving facilities. The liquid level in the liquid chamber fluctuates depending on inlet liquid rates and maximum liquid extraction rate by the liquid pump. Beyond PDG operating envelope, the simulator was able to predict decreasing separation efficiency and chamber liquid level. The simulation results implied that the pressure drop reduction will be higher for longer pipelines which make the technology suitable for long subsea tiebacks.

## Introduction

### Background of Pseudo Dry Gas Technology

Deepwater gas field development with long tieback poses flow assurance challenges due to liquid accumulation, causing high backpressure and extensive compression and pumping energy requirements. This leads to high OPEX, CAPEX, and high CO<sub>2</sub> emissions. A novel Pseudo Dry Gas (PDG) concept has been developed as an inline gas-liquid separator (liquid removal system) which separates liquids into a dedicated liquid line from the gas to induce hydraulically ‘dry gas’ behavior within a wet gas pipeline over the operating envelope.

NEL were contracted to provide high pressure wet-gas flow testing within the Advanced Multiphase Facility (AMF) on the PDG system prototype, to progress the current Technical Readiness Level (TRL) concept; turning a validated concept into a system tested technology. The tests were carried out at nominal pressures ranging from 25 barg to 125 barg and at a temperature of 40 °C. Over four hundred test points were taken during the test program, which ran from 8th of May 2023 and was successfully completed on the 28th of July 2023. A total of more than 1200 testing hours were conducted with the installed PDG system.

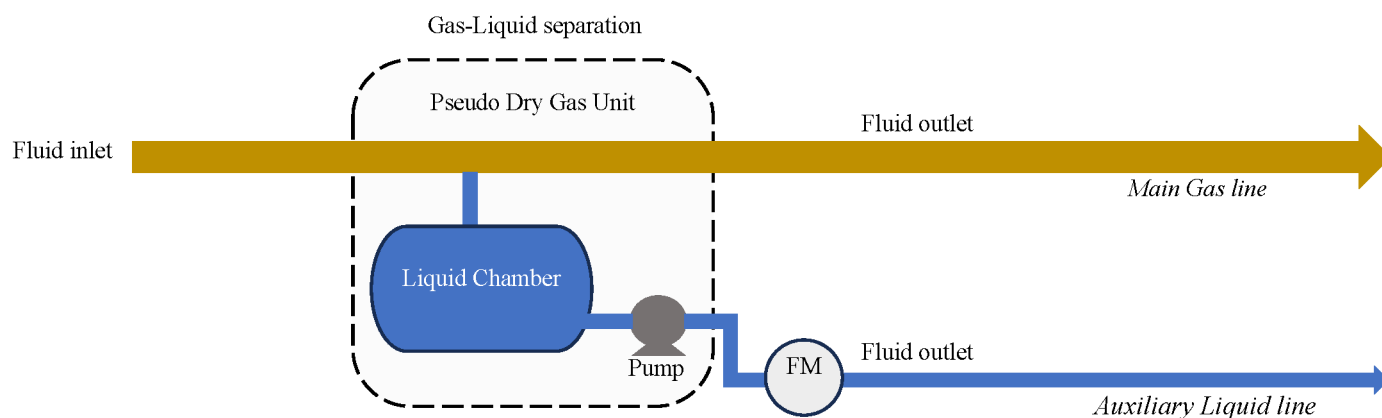


Figure 1—Illustration of PDG unit for a gas pipeline.

## Methodology

### PDG Tool Modeling: The Conceptual Framework

A cloud solution for online flow assurance monitoring has been developed to help operators monitor and optimize their flow assurance mitigation strategies. A commercial transient multiphase flow simulator, FLUX Simulator, has been implemented to gather the field information and to solve the multiphase flow transport equations. The solution can be used for real-time monitoring, look-ahead, and what-if studies. This framework allows development of a user-defined model, SDK API, for additional functionalities other than multiphase flow simulation. In this case, a SDK API for the Pseudo Dry Gas, was developed to calculate in-line gas-liquid separation behavior, i.e., separation efficiency, liquid level in the liquid chamber, pump characteristic, fluid outlet rates in the original gas line and in the auxiliary liquid line based on the inputs as shown in Figure 2. The separation efficiency is calculated based on Froude number while the liquid level is calculated based on mass conservation and pump control strategy. The pump (as part of PDG unit) is

responsible for transporting the separated liquid to the receiving facility via the auxiliary liquid line and maintaining the liquid level in the liquid drum. In case of long tieback, the framework allows adding more PDG unit along the gas line and a booster pump in the liquid line depending on the gas-liquid separation and transport requirements. The inputs required for the SDK API are listed below:

1. Fluid data for composition and fluid properties at various pressures and temperatures.
2. Pipeline information, e.g., dimension, inclination, material, insulation, roughness, pipeline topography.
3. Field operating data as real-time data or user defined inputs.

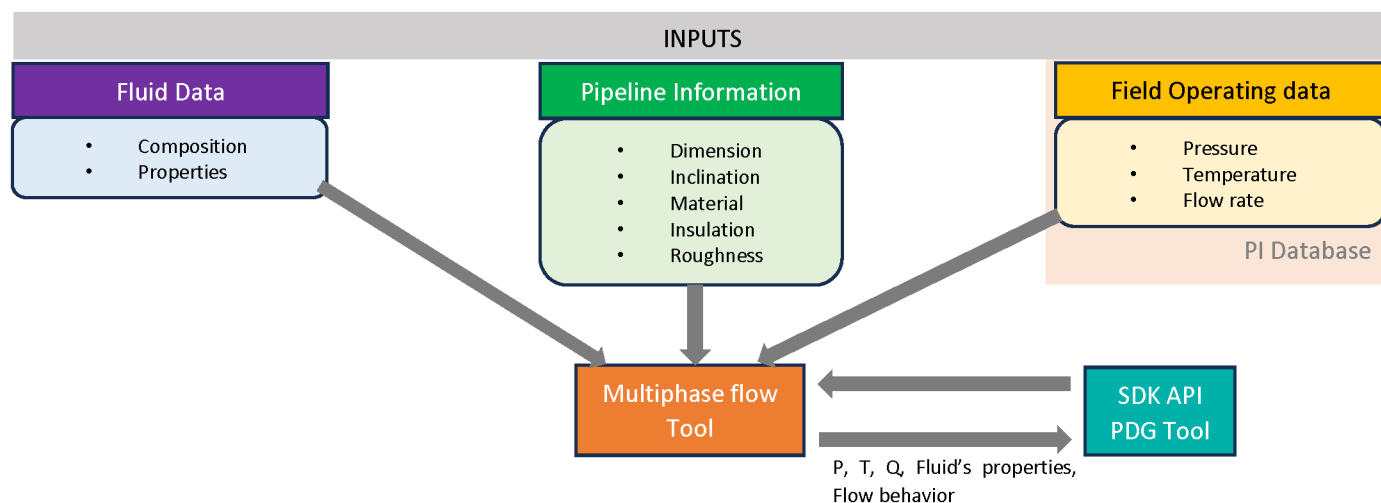


Figure 2—Simplified PDG tool framework.

Three cases are analyzed to test the PDG tool. The first case couples the PDG tool with simplified subsea pipeline. The second case compares the PDG tool results with experimental high-pressure data. Finally, the third case benchmarks with available commercial software.

### Tool Testing with Simplified Subsea Pipeline

Simplified and scaled down subsea pipeline with a PDG unit (as shown in Figure 3) was considered to test the capability and stability of the PDG Tool. The gas and liquid line diameter were set at 6" and 3", respectively. A basic pump control strategy was considered: zero drain rate at 30% liquid level and max. drain rate at and above 70% liquid level with linear increase in between. A transient scenario was considered for the testing, covering various possible scenarios at field, i.e., startup, turndown, shutdown, ramp-up to beyond PDG operating envelope, and normal (constant) production within the operating envelope. The scenario is outlined in

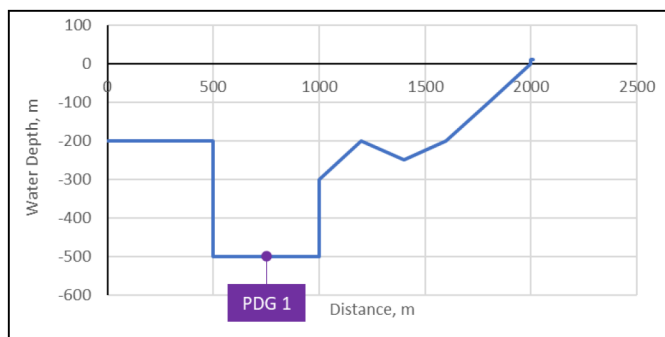


Figure 3—Simplified subsea pipeline for PDG Tool testing.

Table 1. Gas and liquid line arrival pressures were set at 30 and 2 barg, respectively. Fluid composition from a deepwater gas field was used with fluid inlet temperature of 50 °C. The fluid has OGR of 20 bbl/MMScf, and WGR of 7.8 bbl/MMScf. The tool has also undergone successful testing with two PDG units, demonstrating reliable performance. However, the corresponding results are beyond the scope of the current study and will be detailed in a subsequent publication.

Table 1—Transient scenario for PDG Tool testing.

| Time | Rates | Scenario         |
|------|-------|------------------|
| min  | %     |                  |
| 0    | 100%  | Start-up         |
| 60   | 0%    | Ramp-down        |
| 120  | 0%    | Shutdown         |
| 132  | 20%   | Ramp-up          |
| 192  | 20%   | Constant         |
| 480  | 500%  | Ramp-up          |
| 540  | 500%  | Beyond range     |
| 780  | 100%  | Ramp-down        |
| 2220 | 100%  | Normal operation |

### Tool Validation with High Pressure Flow Loop Data

High pressure 6" flow loop test data, conducted at National Engineering Laboratory, Scotland - UK (NEL), with horizontal and near horizontal configurations were used to validate the accuracy of the PDG Tool. The test data covers pressure range of 25 barg to 125 barg at an operating temperature of 40 °C. Various liquids were used during the flow loop test i.e. Exxol D60, Exxol D140, brine water, mixture of Exxol D140 and brine water, with Nitrogen gas as the gas phase. Details about test procedures, test points, test matrix basis, and the data processing can be found in (P. T. Bhaskoro, 2025) (P. Bhaskoro and J. Johar, et al., 2024). The physical built configurations and operating range are presented again in Table 2 and Table 3. The test points are presented in Figure 4.

Table 2—Physical built configurations.

| PDG Configuration       | Liquids                  | Gas      |
|-------------------------|--------------------------|----------|
| Flat – With PDG         | Exxol D140, Saline Water | Nitrogen |
| Flat – Without PDG      | Exxol D140, Saline Water | Nitrogen |
| Upward 2° – Without PDG | Exxol D140, Saline Water | Nitrogen |
| Upward 2° – Without PDG | Exxol D60                | Nitrogen |
| Upward 2° – With PDG    | Exxol D140, Saline Water | Nitrogen |
| Upward 2° – With PDG    | Exxol D60                | Nitrogen |
| Flat – With PDG         | Exxol D60                | Nitrogen |
| Flat – Without PDG      | Exxol D60                | Nitrogen |

Table 3—Testing parameter range considered in the flow loop test.

| Test Parameter                          | Range                    |
|-----------------------------------------|--------------------------|
| Inclination Angle                       | 0 and 2 degrees (Upward) |
| Pressure                                | 25 – 125 barg            |
| Liquid Superficial Velocity (Usl) Range | 0.005 – 0.3 m/s          |
| Gas Superficial Velocity (Usg) Range    | 2 – 10 m/s               |

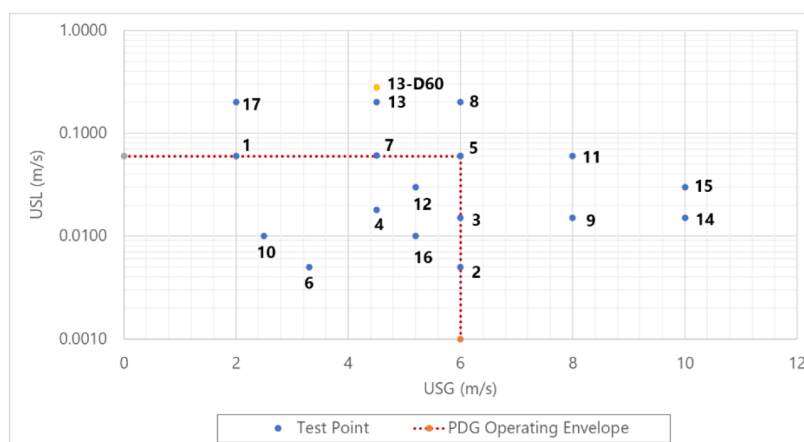


Figure 4—Pseudo Dry Gas testing points in high pressure flow loop.

## Benchmarking with Commercial Software

A deepwater 140-km gas pipeline was used for benchmarking with PIPESIM and OLGA simulators for steady state case. The pipeline inlet is at water depth of 2000 m and has a nominal diameter of 24 inches. Partial burial is modelled by including a 21.22 mm soil layer outside the steel pipe. Ambient conditions are specified to vary linearly with seawater depth from 27 °C at the surface to 3 °C at a depth of 2000 m, and the seawater current is set to 0.5 m/s. The pipeline model trajectory is presented in Figure 5.

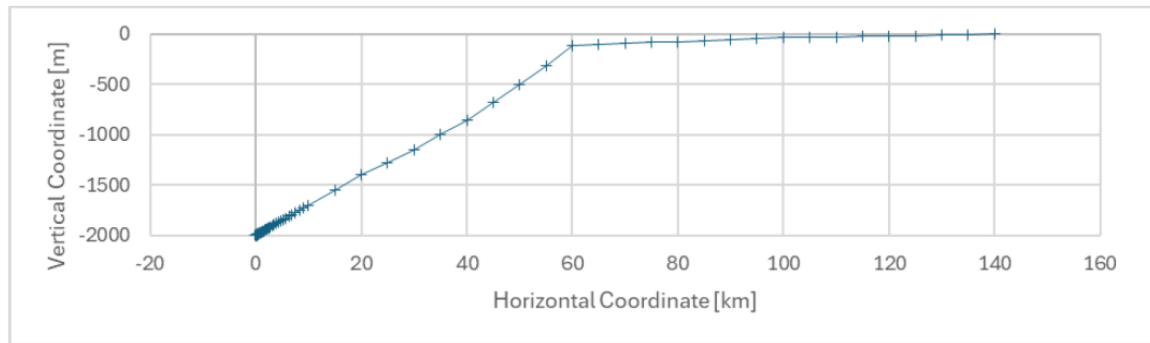


Figure 5—Deepwater pipeline model trajectory for benchmarking.

Three fluid sources are located at the inlet of the model, where one represents water-saturated (at 750 psi, 75 °C) reservoir gas and associated condensate (2 STB/MMscf). Another source accounts for produced water, and the last source represents injected MEG (hydrate inhibitor). The MEG source is modelled as pure water with no adjustments for viscosity and density of the inhibitor. The PDG separator is positioned at the boundary between the short horizontal pipe segment and the first inclined pipe. Separated liquids are removed from the model. A fixed gas line outlet pressure of 64 barg is applied.

## Results and Discussions

### PDG Tool Testing with Simplified Subsea Pipeline

Simulation results of transient inlet fluid rates and the corresponding PDG's separation efficiency are presented in Figure 6. Figure 6a shows the gas, oil and water rates at the inlet of PDG device. The liquid and gas surges observed between the 3rd and 8th hour of simulation correspond to a production ramp-up from 20% to 500% of the normal operating rate. Subsequently, between the 9th and 13th hour, the liquid and gas rates were gradually ramped down, returning to normal production levels. Figure 6b, shows the decreasing separation efficiency as the fluid is ramped up beyond PDG operating envelope. Minimum separation efficiency is reached due to high  $U_{sg}$ , which qualitatively follows the flow loop results. The efficiency increases back when the flow is ramped down toward PDG operating envelope which eventually reaches maximum efficiency. The separation efficiency maintains at maximum efficiency during normal production which is within the envelope.

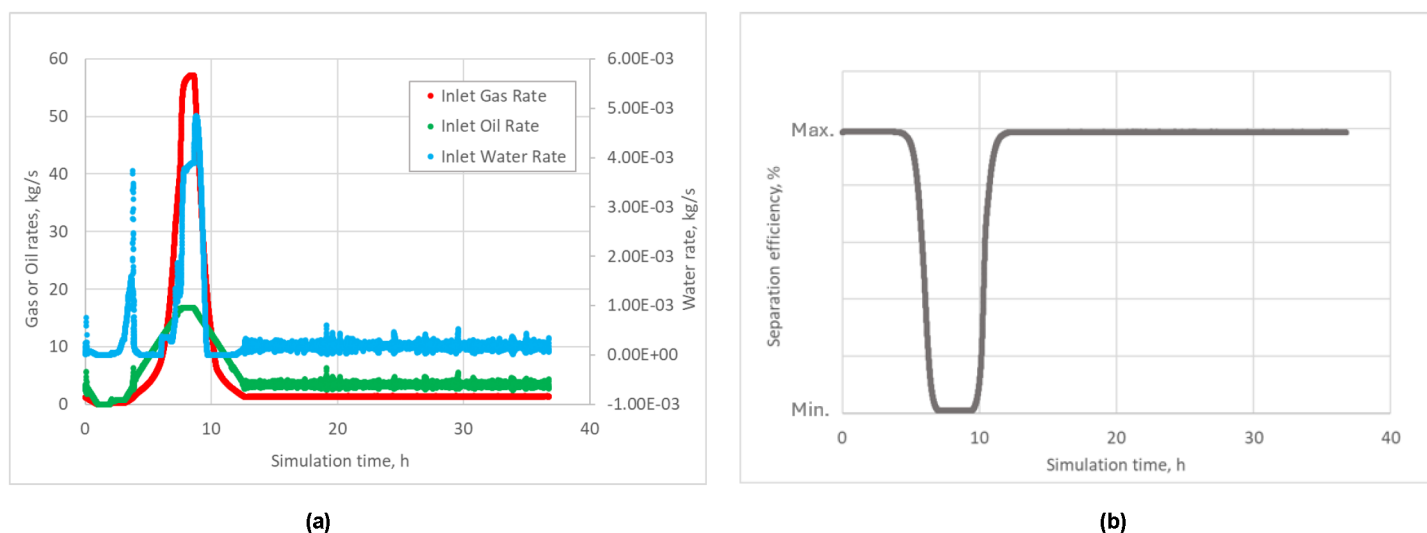


Figure 6—PDG Tool simulation results, a) PDG Inlet fluids rates, and b) PDG separation efficiency.

Figure 7 presents the predicted oil and water flow rates, both separated from the main gas line and discharged from the liquid chamber. The separated oil and water rates refer to the portion of liquid diverted from the main gas stream into the liquid chamber. The outlet oil and water rates represent the flow exiting the liquid chamber, driven by the action of the pump. The separated liquid experiences surge due to ramp-up and ramp-down events in the flow rates. The maximum outlet liquid rate occurs twice (at 5 and 12 hours) corresponding to pump ramp-up and ramp-down events, which increase and decrease the outlet oil and water rates.

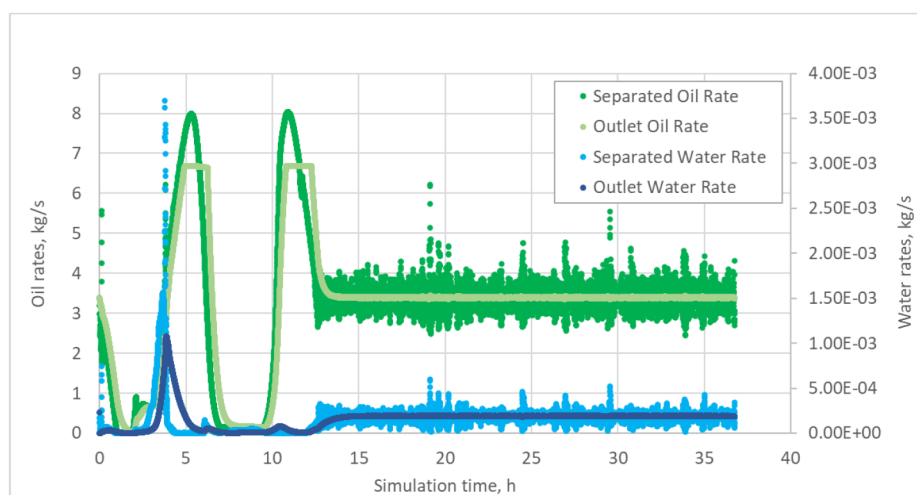


Figure 7—Simulation results of liquid separate rates in PDG unit.

The first peak is linked to the inlet oil ramp-up at 3 hours. The sudden increase in separated oil raises the liquid chamber level, as shown in Figure 8, triggering a pump ramp-up and increasing the outlet oil flow rate to the pump's maximum capacity at 4.5 hours. This maximum output rate is maintained until the chamber's liquid level begins to decline.

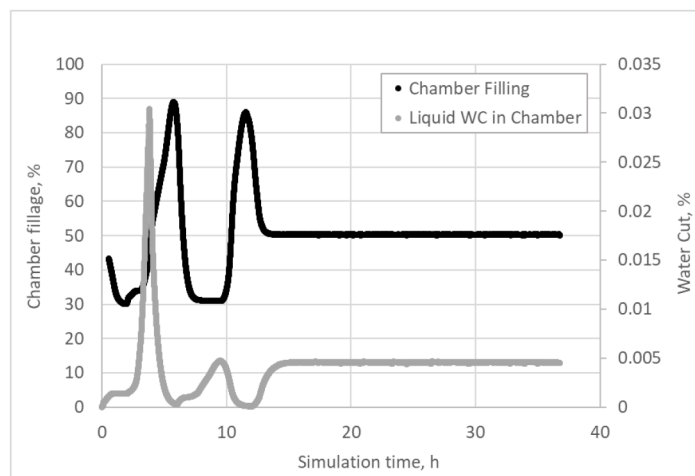


Figure 8—Simulation results of liquid volume in the liquid chamber and the water cut.

After 8 hours of simulation, the inlet gas flow rate reaches its peak, while the separation efficiency drops to its minimum. This results sharply decrease in both the separated and outlet oil rates. Eventually, the gas flow rate ramps down to the PDG envelope, allowing the separation efficiency to recover.

The second peak is associated with the inlet gas ramp-down at 10 hours, which brings the flow rate within the PDG envelope and improves separation efficiency. This behavior increases the separated liquid rate. The rise in separated oil again elevates the liquid chamber level, prompting another pump ramp-up and boosting the outlet oil flow rate to its maximum capacity at 11 hours. This rate is sustained until the chamber's liquid level begins to fall from its peak to eventually reach the set point and stay constant during normal production.

The results in Figure 9 show that the PDG can keep liquid hold up to be minimal along the gas line (mostly in stratified flow regime), minimizing slugging and liquid surge to receiving facilities. High liquid holdup at several locations downstream of PDG is due to pipeline topography which creates a V-shape configuration trapping the liquid.

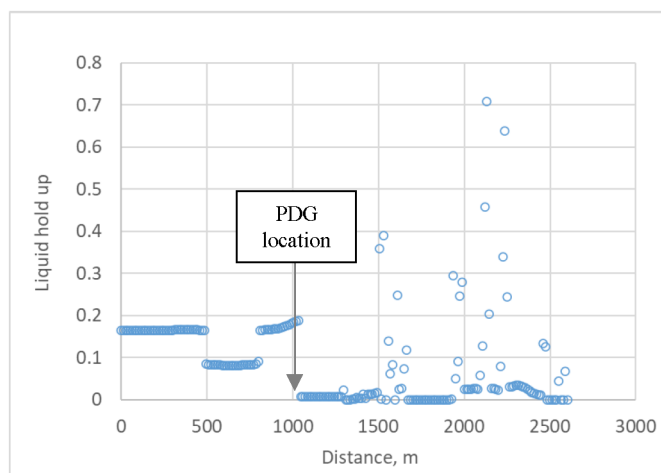


Figure 9—Simulation results of liquid hold up along the main gas line.

### Validation Results with High Pressure Flow Loop Data

Gas-liquid separation efficiency prediction error for N<sub>2</sub>-D60 separation efficiency with horizontal and 2° upward configuration is presented in Figure 10. The prediction errors are categorized into three groups:



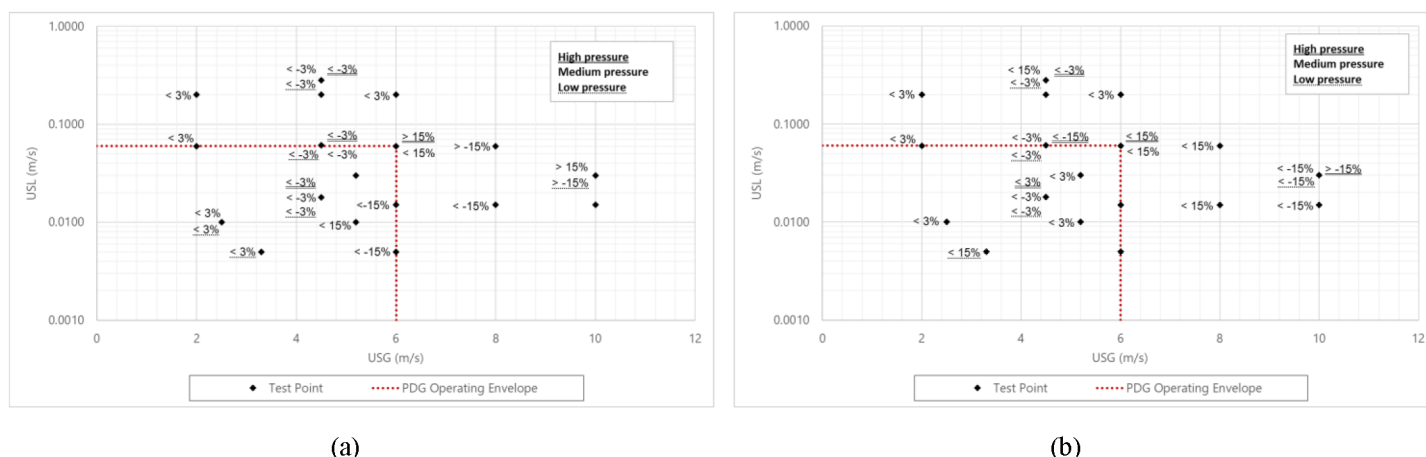


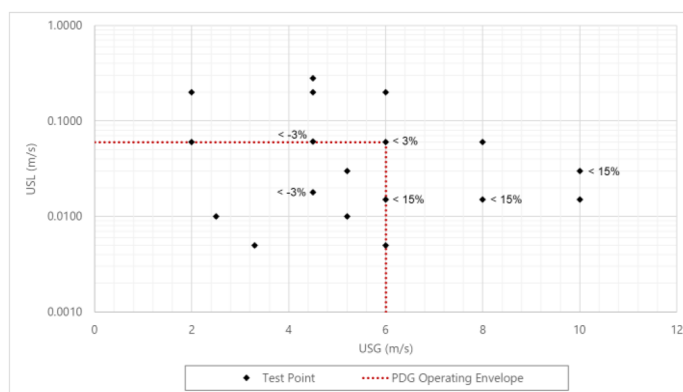
Figure 10—Prediction error based on flow loop data for N<sub>2</sub>-D60 multiphase flow, a). With horizontal configuration, and b). with 2° upward configuration.

1. Minor errors – over-prediction less than 3% and under-prediction better than -3%.
2. Moderate errors – over-prediction between 3% and 15%, and under-prediction between -3% and -15%.
3. Significant errors – over-prediction greater than 15% and under-prediction worse than -15%.

With horizontal configuration, the results show that the prediction is considered good, with errors less than 15%, when superficial gas (USG) and superficial liquid (USL) velocity below 6 m/s and 0.2 m/s, respectively. For slightly upward configuration, the good prediction is extended up to 10 m/s of superficial gas velocity. The trend shows that the error is higher with higher USG. It could be due to flow uncertainty caused by liquid entrainment and higher USG. The flow uncertainty makes it difficult to reach a steady state flow, especially for high USG, and produces fluctuation during measurement. Higher measurement error due to low liquid flow rate also contributes to the prediction error. At high pressure, more gas will diffuse into the liquid, reducing the density, and making liquid entrainment more favorable. It induces higher flow uncertainty and hence, higher prediction error.

Prediction error with D140 as the liquid phase, at medium pressure, for both horizontal and upward configurations, are presented in Figure 11. Like D60 case, the prediction error is good within the same USG and USL range. Higher prediction error is observed for higher USG which is believed to be due to the same reason. Nevertheless, comparison between D140 and D60 cases shows higher prediction error for D140 for upward configuration at high USG. It could be due to higher density of D140 than D60, which may cause backflow. Considering that liquid backflow will affect gas-liquid separation efficiency and the fact that it could not be measured during flow loop test, liquid backflow causes measurement error which may be responsible for the high prediction error. For horizontal configuration, the opposite situation is observed, where the prediction error for D140 at high USG is significantly lower than D60 case. It can be explained from the fluid's density. Since D140's density is higher than D60, the flow is calmer (less flow uncertainty) for D140 than for D60, leading to more accurate measurement and subsequently lower prediction error.

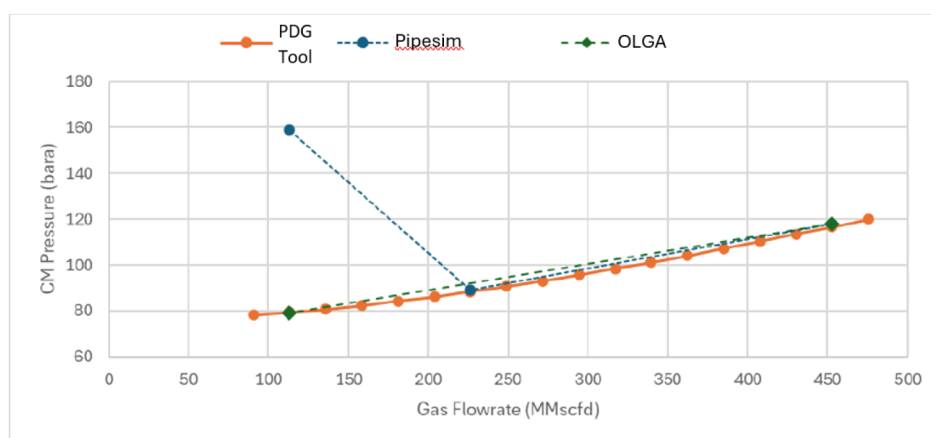




**Figure 13—Prediction error based on flow loop data for  $N_2$ -D140/Brine water mixture, multiphase flow, with horizontal configuration**

### PDG Tool Benchmarking with Commercial Simulators

The steady state simulation results for a 140 km deepwater pipeline with a PDG unit, in terms of pressure at central manifold (CM) and total liquid content in the pipeline, are in excellent agreement with OLGA and Pipesim results, as shown in Figure 14 and Figure 15. When considering the graph with logarithmic scale for liquid content, there may appear to be a disagreement between the PDG Tool and OLGA results at low flow rates. This deviation is likely caused by the fact that the flow models are not solved in the same way. Given that the total volume of the pipeline is around 35000 m<sup>3</sup>, the difference between liquid content predictions (less than 30 m<sup>3</sup>) is believed to be within the expected accuracy of both simulators.



**Figure 14—Comparison of pressure results between PIPESIM, OLGA and PDG Tool at various inlet rates**

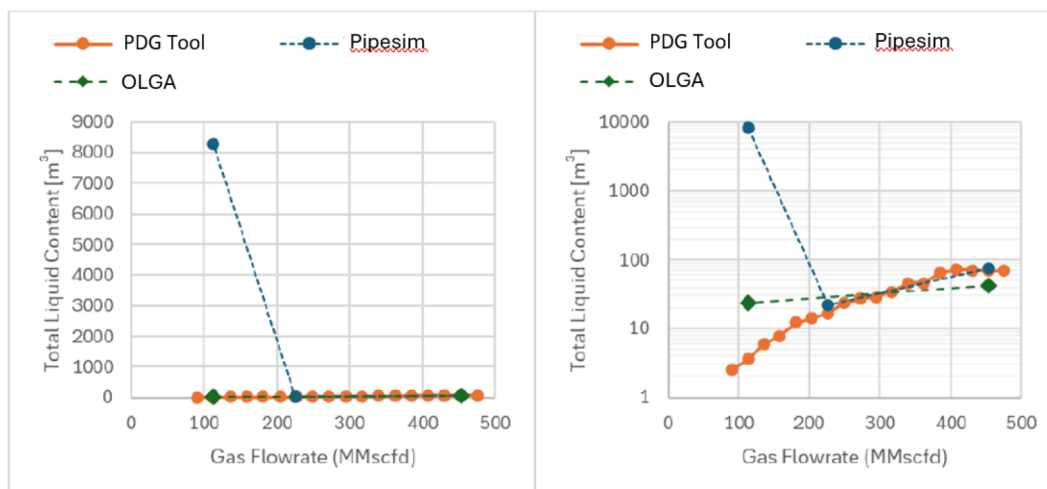


Figure 15—Comparison of liquid content results between PIPESIM, OLGA and FLUX Simulator at various inlet rates. Linear scale to the left, logarithmic scale to the right

### Field Deployment, Scalability, and Real-Time Integrations

The monitoring tool developed for Pseudo Dry Gas (PDG) technology is fully prepared for testing/deployment in the field, contingent upon the installation of a PDG unit. The simulation engine demonstrates a latency of less than 10 seconds, ensuring near real-time responsiveness. Furthermore, the tool offers configurable update intervals, with input and output data refresh rates adjustable from 30 seconds onward, depending on operational requirements and system integration constraints.

The simulation tool has also been validated under transient operational scenarios—including ramp-up, turndown, startup, and shutdown—using a deepwater pipeline (2,000m of water depth) configuration with a long tieback distance of 140 km. Detailed results of these simulations can be found in (P. T. Bhaskoro, 2025), which include:

- Total liquid arrival rate at the onshore facility during and post ramp-up operations
- Required inlet pressure during and following ramp-up
- Calculated pressure profiles at both pipeline ends during shut-in conditions
- Liquid arrival rate during and after startup following shut-in from design flow rate
- Required inlet pressure during and after startup following shut-in
- PDG separation efficiency under transient conditions: (a) during and post ramp-up, (b) during and post restart

Finally, the techno-economic analysis of PDG technology application for a deepwater gas field with long tieback showed that the technology able to reduce the CAPEX and OPEX by 14.7% and 20.5%, respectively, in comparison with gas Floating Production Storage Operation (FPU) concept. The details can be found in (A. Azuhan, 2024).

### Conclusions

The PDG Online Monitoring Tool has been successfully developed to predict the separation performance of the Pseudo Dry Gas system and its impact on the main gas and liquid flowlines, with acceptable accuracy. When integrated with field sensor data, it enables digital twin capabilities for real-time online monitoring. Simulation results confirm the tool's ability to handle transient scenarios both within the PDG operating envelope – achieving high separation efficiency – and outside it, where efficiency is reduced. The simulator

effectively models the influence of multiphase flow dynamics on separation performance under varying conditions. Furthermore, the PDG Tool provides detailed insights into the dynamic behavior within the liquid chamber, influenced by upstream multiphase flow. It can simulate separation efficiency, pump control strategies, and predict key parameters such as liquid accumulation (chamber liquid level) and water cut. This capability is essential for modeling pipelines equipped with PDG systems, especially when addressing complex flow assurance challenges such as wax and hydrate formation.

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